

B. STAAD INPUT DATA

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*****
*          STAAD.Pro          *
*          Version 2007      Build 01          *
*          Proprietary Program of              *
*          Research Engineers, Intl.            *
*          Date=    OCT 14, 2008                *
*          Time=    15:48: 6                    *
*          USER ID: ENGINEERING                 *
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1. STAAD SPACE
INPUT FILE: SKYLIGHT-GLASS_1975 X 1933_A1.STD
2. START JOB INFORMATION
3. JOB NAME GLASS ANALYSIS - TYPICAL DIMENSION
4. JOB CLIENT DMP
5. JOB COMMENT FINAL
6. ENGINEER NAME AA
7. JOB REV 2
8. ENGINEER DATE 17-MAR-08
9. JOB NO 101
10. END JOB INFORMATION
11. INPUT WIDTH 79
12. UNIT MMS NEWTON
13. JOINT COORDINATES
14. 1 0 0 0; 2 1975 0 0; 3 0 0 1933; 4 1975 0 1933; 5 0 0 115; 6 1975 0 115
15. 7 0 0 1818; 8 1975 0 1818; 9 115 0 0; 10 115 0 1933; 11 115 0 115
16. 12 115 0 1818; 13 1860 0 0; 14 1860 0 1933; 15 1860 0 115; 16 1860 0 1818
17. 17 115 0 1704.47; 18 0 0 1704.47; 19 115 0 1590.93; 20 0 0 1590.93
18. 21 115 0 1477.4; 22 0 0 1477.4; 23 115 0 1363.87; 24 0 0 1363.87
19. 25 115 0 1250.33; 26 0 0 1250.33; 27 115 0 1136.8; 28 0 0 1136.8
20. 29 115 0 1023.27; 30 0 0 1023.27; 31 115 0 909.733; 32 0 0 909.733
21. 33 115 0 796.2; 34 0 0 796.2; 35 115 0 682.667; 36 0 0 682.667
22. 37 115 0 569.133; 38 0 0 569.133; 39 115 0 455.6; 40 0 0 455.6
23. 41 115 0 342.067; 42 0 0 342.067; 43 115 0 228.533; 44 0 0 228.533
24. 45 1975 0 1704.47; 46 1860 0 1704.47; 47 1975 0 1590.93; 48 1860 0 1590.93
25. 49 1975 0 1477.4; 50 1860 0 1477.4; 51 1975 0 1363.87; 52 1860 0 1363.87
26. 53 1975 0 1250.33; 54 1860 0 1250.33; 55 1975 0 1136.8; 56 1860 0 1136.8
27. 57 1975 0 1023.27; 58 1860 0 1023.27; 59 1975 0 909.733; 60 1860 0 909.733
28. 61 1975 0 796.2; 62 1860 0 796.2; 63 1975 0 682.667; 64 1860 0 682.667
29. 65 1975 0 569.133; 66 1860 0 569.133; 67 1975 0 455.6; 68 1860 0 455.6
30. 69 1975 0 342.067; 70 1860 0 342.067; 71 1975 0 228.533; 72 1860 0 228.533
31. 73 1743.67 0 1818; 74 1743.67 0 1933; 75 1627.33 0 1818; 76 1627.33 0 1933
32. 77 1511 0 1818; 78 1511 0 1933; 79 1394.67 0 1818; 80 1394.67 0 1933
33. 81 1278.33 0 1818; 82 1278.33 0 1933; 83 1162 0 1818; 84 1162 0 1933
34. 85 1045.67 0 1818; 86 1045.67 0 1933; 87 929.333 0 1818; 88 929.333 0 1933
35. 89 813 0 1818; 90 813 0 1933; 91 696.667 0 1818; 92 696.667 0 1933
36. 93 580.333 0 1818; 94 580.333 0 1933; 95 464 0 1818; 96 464 0 1933
37. 97 347.667 0 1818; 98 347.667 0 1933; 99 231.333 0 1818; 100 231.333 0 1933
38. 101 231.333 0 115; 102 231.333 0 0; 103 347.667 0 115; 104 347.667 0 0
39. 105 464 0 115; 106 464 0 0; 107 580.333 0 115; 108 580.333 0 0
40. 109 696.667 0 115; 110 696.667 0 0; 111 813 0 115; 112 813 0 0
41. 113 929.333 0 115; 114 929.333 0 0; 115 1045.67 0 115; 116 1045.67 0 0
42. 117 1162 0 115; 118 1162 0 0; 119 1278.33 0 115; 120 1278.33 0 0
43. 121 1394.67 0 115; 122 1394.67 0 0; 123 1511 0 115; 124 1511 0 0
44. 125 1627.33 0 115; 126 1627.33 0 0; 127 1743.67 0 115; 128 1743.67 0 0
45. 129 231.333 0 1704.47; 130 347.667 0 1704.47; 131 464 0 1704.47
46. 132 580.333 0 1704.47; 133 696.667 0 1704.47; 134 813 0 1704.47
47. 135 929.333 0 1704.47; 136 1045.67 0 1704.47; 137 1162 0 1704.47
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COMPLETE JOINT COORDINATES WILL BE PROVIDED IN SOFT COPY UPON REQUEST.

108. ELEMENT INCIDENCES SHELL

109. 1 5 11 9 1; 2 15 6 2 13; 3 14 4 8 16; 4 3 10 12 7; 5 7 12 17 18; 6 18 17 19 20
110. 7 20 19 21 22; 8 22 21 23 24; 9 24 23 25 26; 10 26 25 27 28; 11 28 27 29 30
111. 12 30 29 31 32; 13 32 31 33 34; 14 34 33 35 36; 15 36 35 37 38; 16 38 37 39 40
112. 17 40 39 41 42; 18 42 41 43 44; 19 44 43 11 5; 20 16 8 45 46; 21 46 45 47 48
113. 22 48 47 49 50; 23 50 49 51 52; 24 52 51 53 54; 25 54 53 55 56; 26 56 55 57 58
114. 27 58 57 59 60; 28 60 59 61 62; 29 62 61 63 64; 30 64 63 65 66; 31 66 65 67 68
115. 32 68 67 69 70; 33 70 69 71 72; 34 72 71 6 15; 35 16 73 74 14; 36 73 75 76 74
116. 37 75 77 78 76; 38 77 79 80 78; 39 79 81 82 80; 40 81 83 84 82; 41 83 85 86 84
117. 42 85 87 88 86; 43 87 89 90 88; 44 89 91 92 90; 45 91 93 94 92; 46 93 95 96 94
118. 47 95 97 98 96; 48 97 99 100 98; 49 99 12 10 100; 50 11 101 102 9
119. 51 101 103 104 102; 52 103 105 106 104; 53 105 107 108 106; 54 107 109 110 108
120. 55 109 111 112 110; 56 111 113 114 112; 57 113 115 116 114; 58 115 117 118 116
121. 59 117 119 120 118; 60 119 121 122 120; 61 121 123 124 122; 62 123 125 126 124
122. 63 125 127 128 126; 64 127 15 13 128; 65 12 99 129 17; 66 99 97 130 129
123. 67 97 95 131 130; 68 95 93 132 131; 69 93 91 133 132; 70 91 89 134 133
124. 71 89 87 135 134; 72 87 85 136 135; 73 85 83 137 136; 74 83 81 138 137
125. 75 81 79 139 138; 76 79 77 140 139; 77 77 75 141 140; 78 75 73 142 141
126. 79 73 16 46 142; 80 17 129 143 19; 81 129 130 144 143; 82 130 131 145 144
127. 83 131 132 146 145; 84 132 133 147 146; 85 133 134 148 147; 86 134 135 149 148
128. 87 135 136 150 149; 88 136 137 151 150; 89 137 138 152 151; 90 138 139 153 152
129. 91 139 140 154 153; 92 140 141 155 154; 93 141 142 156 155; 94 142 46 48 156
130. 95 19 143 157 21; 96 143 144 158 157; 97 144 145 159 158; 98 145 146 160 159
131. 99 146 147 161 160; 100 147 148 162 161; 101 148 149 163 162
132. 102 149 150 164 163; 103 150 151 165 164; 104 151 152 166 165
133. 105 152 153 167 166; 106 153 154 168 167; 107 154 155 169 168
134. 108 155 156 170 169; 109 156 48 50 170; 110 21 157 171 23; 111 157 158 172 171
135. 112 158 159 173 172; 113 159 160 174 173; 114 160 161 175 174
136. 115 161 162 176 175; 116 162 163 177 176; 117 163 164 178 177
137. 118 164 165 179 178; 119 165 166 180 179; 120 166 167 181 180
138. 121 167 168 182 181; 122 168 169 183 182; 123 169 170 184 183
139. 124 170 50 52 184; 125 23 171 185 25; 126 171 172 186 185; 127 172 173 187 186
140. 128 173 174 188 187; 129 174 175 189 188; 130 175 176 190 189
141. 131 176 177 191 190; 132 177 178 192 191; 133 178 179 193 192
142. 134 179 180 194 193; 135 180 181 195 194; 136 181 182 196 195
143. 137 182 183 197 196; 138 183 184 198 197; 139 184 52 54 198; 140 25 185 199 27
144. 141 185 186 200 199; 142 186 187 201 200; 143 187 188 202 201
145. 144 188 189 203 202; 145 189 190 204 203; 146 190 191 205 204
146. 147 191 192 206 205; 148 192 193 207 206; 149 193 194 208 207
147. 150 194 195 209 208; 151 195 196 210 209; 152 196 197 211 210
148. 153 197 198 212 211; 154 198 54 56 212; 155 27 199 213 29; 156 199 200 214 213
149. 157 200 201 215 214; 158 201 202 216 215; 159 202 203 217 216
150. 160 203 204 218 217; 161 204 205 219 218; 162 205 206 220 219
151. 163 206 207 221 220; 164 207 208 222 221; 165 208 209 223 222
152. 166 209 210 224 223; 167 210 211 225 224; 168 211 212 226 225
153. 169 212 56 58 226; 170 29 213 227 31; 171 213 214 228 227; 172 214 215 229 228
154. 173 215 216 230 229; 174 216 217 231 230; 175 217 218 232 231
155. 176 218 219 233 232; 177 219 220 234 233; 178 220 221 235 234
156. 179 221 222 236 235; 180 222 223 237 236; 181 223 224 238 237
157. 182 224 225 239 238; 183 225 226 240 239; 184 226 58 60 240; 185 31 227 241 33
158. 186 227 228 242 241; 187 228 229 243 242; 188 229 230 244 243
159. 189 230 231 245 244; 190 231 232 246 245; 191 232 233 247 246
160. 192 233 234 248 247; 193 234 235 249 248; 194 235 236 250 249
161. 195 236 237 251 250; 196 237 238 252 251; 197 238 239 253 252
162. 198 239 240 254 253; 199 240 60 62 254; 200 33 241 255 35; 201 241 242 256 255
163. 202 242 243 257 256; 203 243 244 258 257; 204 244 245 259 258
164. 205 245 246 260 259; 206 246 247 261 260; 207 247 248 262 261
165. 208 248 249 263 262; 209 249 250 264 263; 210 250 251 265 264
166. 211 251 252 266 265; 212 252 253 267 266; 213 253 254 268 267
167. 214 254 62 64 268; 215 35 255 269 37; 216 255 256 270 269; 217 256 257 271 270

COMPLETE ELEMENT INCIDENCES SHELL WILL BE PROVIDED IN SOFT COPY UPON REQUEST.

191. ELEMENT PROPERTY
192. 1 TO 289 THICKNESS 24
193. DEFINE MATERIAL START
194. ISOTROPIC GLASS
195. E 72000
196. POISSON 0.22
197. DENSITY 2.4525E-005
198. ALPHA 8.5E-006
199. DAMP 0.03
200. END DEFINE MATERIAL
201. CONSTANTS
202. MATERIAL GLASS ALL
203. SUPPORTS
204. 11 PINNED
205. 15 FIXED BUT FX MX MZ
206. 12 FIXED BUT FZ MX MY MZ
207. 16 FIXED BUT FX FZ MX MY MZ
208. LOAD 1 LOADTYPE MASS TITLE SELFWEIGHT
209. SELFWEIGHT Y -1
210. LOAD 2 LOADTYPE LIVE TITLE CASE ONE_POINT LOAD
211. JOINT LOAD
212. 143 149 155 213 219 225 283 289 295 FY -500
213. LOAD 3 LOADTYPE LIVE TITLE CASE TWO_PLATE LOAD
214. ELEMENT LOAD
215. 1 TO 289 PR GY -0.001
216. LOAD 4 LOADTYPE TEMPERATURE TITLE THERMAL LOAD
217. TEMPERATURE LOAD
218. 1 TO 289 TEMP 0 32 32
219. LOAD 5 LOADTYPE WIND TITLE WIND LOAD_POS
220. ELEMENT LOAD
221. 1 TO 289 PR GY 0.00152
222. LOAD 6 LOADTYPE WIND TITLE WIND LOAD_NEG
223. ELEMENT LOAD
224. 1 TO 289 PR GY -0.00292
225. LOAD COMB 7 (SLF + POINT LOAD)_NOMINAL
226. 1 1.0 2 1.0
227. LOAD COMB 8 (SLF + PLATE LOAD)_NOMINAL
228. 1 1.0 3 1.0
229. LOAD COMB 9 (SLF + POS_WL)_NOMINAL
230. 1 1.0 5 1.0
231. LOAD COMB 10 (SLF + NEG_WL)_NOMINAL
232. 1 1.0 6 1.0
233. LOAD COMB 11 (SLF + POINT LOAD + THERMAL LOAD)_FACTORED
234. 1 1.4 2 1.4 4 1.0
235. LOAD COMB 12 (SLF + PLATE LOAD + THERMAL LOAD)_FACTORED
236. 1 1.4 3 1.4 4 1.0
237. LOAD COMB 13 (SLF + WIND LOAD_POS + THERMAL LOAD)_FACTORED
238. 1 1.4 5 1.4 4 1.0
239. LOAD COMB 14 (SLF + WIND LOAD_NEG + THERMAL LOAD)_FACTORED
240. 1 1.4 6 1.4 4 1.0
241. PERFORM ANALYSIS